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Yr/Sec: BSIT WMT 2-2

W1-Quiz1 Activity:

Introduction to Algorithms

A. Discuss Briefly:

What is an algorithm, and how is it related to the tasks you perform in your daily life?

An algorithm is a set of step-by-step instructions that is used to complete a task. I probably used an algorithms in my daily life. A perfect everyday example of an algorithm that matches how I process information is following a recipe to cook a meal. Just like a computer algorithm, a recipe relies on the exact same core structure: Input ➡ Process ➡ Output.

• Input: raw ingredients (e.g., rice, egg, garlic, and oil for fried rice).

• Process (The Algorithm): step-by-step instructions I must follow in order.

• Output: A finished plate of food.

Why is it important for an IT student to learn algorithms?

Algorithms are important for an IT student because they teach us how to solve problems in a clear and organized way. In this regard, learning algorithms can helps us understand what steps we need to follow to make a program work properly. Additionally, through learning algorithms, we can also create programs that are faster, more efficient, and easier to understand. As an IT student, learning algorithms is important because it improves our problem-solving and logical thinking skills, which we can use not only in programming but also in other areas of technology.

B. **Convert each JavaScript code into a step-by-step algorithm using simple English or pseudocode.**

Guide: Describe what the program does from the beginning until the loop stops. You may use numbered steps.

1.

```
for (let i = 1; i < 10; i *= 2) {
   console.log(i);
}
```

2.

```
for (let i = 0; i < 10; i++) {
   console.log(i);
}
```

First JavaScript Code

1. Set I to 1.
2. Check if I is less than 10.
3. If it is, print the value of i.
4. Multiply I by 2.
5. Check the condition again.
6. The values printed are 1, 2, 4, and 8.
7. After 8, I becomes 16, which is not less than 10, so the loop stops.

Second JavaScript Code

1. Set I to 0.
2. Check if I is less than 10.
3. If it is, print the value of i.
4. Increase I by 1.
5. Repeat the process until I becomes 10.
6. When I is 10, the condition is false, so the loop stops.

C. **Run both JavaScript codes. Based on the outputs and execution of the two codes, answer the following:**

1. **What did you observe about the sequence of numbers produced by the first code?**
   - The numbers are 1, 2, 4, and 8. The value doubles each time.
2. **What did you observe about the sequence of numbers produced by the second code?**
   - The numbers are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. The value increases by 1 each time.
3. **What is the main difference between the way the two loops progress?**
   - The first loop, (I *= 2) multiplies the value by 2 each time, while the second loop, (i++) adds 1 each time.
4. **How does the way an algorithm performs its steps affect its output?**
   - The way the algorithm updates its value affects the numbers it produces. Since the first loop multiplies by 2 and the second loop adds 1, they produce different sequences of numbers.